Sentiment-Driven Portfolio Optimization: Do Headlines Pay After Costs?

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Abstract

A portfolio manager of a weekly rebalanced, long-only U.S. and euro-area equity mandate must decide whether news sentiment is worth adding to the allocation, and how. This study tests two uses of Google News headline sentiment, measured by FinBERT, in a cost-aware Black–Litterman optimiser: as a correction of a price forecast and as the only source of views. It scores 661,986 company headlines for the members of the S&P 100 and EURO STOXX 50 and, following CRISP-DM, selects all models on 2012–2014 before a single test on 2015–2025 with 10 basis points per trade. Sentiment signals mostly followed price moves and showed no lasting continuation. The correction did not improve forecast accuracy but raised net return by +1.92 percentage points a year (95% interval -1.47 to 5.00), and the gain came from heavily covered firms. The sentiment-driven portfolio had a Sharpe ratio of 0.64 against 0.63 for equal weighting, with a slightly lower return, and this edge disappeared when repeated headlines were removed. No difference was statistically significant. The headlines were retrieved in 2026, so the archive may favour stories that stayed online or later proved important; a check against dated U.S. headlines found no sign of this. For a portfolio manager, headline sentiment is useful as a check on a price-based allocation and weak as a stand-alone source of alpha.

Keywords: portfolio optimization; headline sentiment; Google News; FinBERT; Black–Litterman; walk-forward backtest; asset management

1. Introduction

1.1. Background

Investors and analysts follow company news closely. Language models such as FinBERT now turn thousands of headlines a week into tone scores at low cost [1, 2]. It is less clear whether this helps a portfolio after costs: under semi-strong efficiency public news is priced quickly [3], while behavioural finance argues that investors under- and over-react, so tone can move prices beyond the facts reported [4, 5]. Inspired by work that feeds sentiment into allocation through reinforcement learning [6] and the Black–Litterman model [7], this study asks whether a transparent, cost-aware optimiser can let sentiment steer a realistic long-only portfolio.

1.2. Problem statement

The stakeholder is a portfolio manager of a weekly rebalanced, long-only mandate in U.S. and euro-area large caps, who must decide whether a headline-sentiment layer is worth its data, computing and trading costs, and how it should enter the allocation: as a check that confirms or contradicts an existing price-based forecast, or as the driver of the allocation itself. The decision variable is the weekly weight of each stock. The key performance indicators (KPIs) are net return, Sharpe ratio, maximum drawdown and trading cost, against equal weighting, minimum variance and MSCI World. The universe is the 150 members of the S&P 100 and EURO STOXX 50 at the end of 2014 (141 with usable prices), with weekly decisions from 2015 to 2025.

1.3. Research questions

Main question: *How can news sentiment be used in portfolio optimisation to improve the allocation of a weekly, long-only U.S. and euro-area equity portfolio after trading costs?*

Here, news sentiment means *Google News headline sentiment measured by FinBERT*, called sentiment below.

RQ1 Does headline sentiment carry information about future returns that prices have not yet absorbed, and does its effect persist or reverse?

RQ2 Does an allocation in which sentiment confirms or contradicts a price-based forecast improve

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return, risk and drawdown compared with the price-based allocation alone?

RQ3 Does a sentiment-driven allocation outperform equal weighting, minimum variance and the market index after costs, and how robust is the result to costs, timing and market conditions?

1.4. Related work and knowledge gap

The closest studies (Table 1) shape the design. Tone is measured against each firm's own history [8], aggregated weekly [9] and used only from the next session, with a delay test [10]. Reported reversals [5, 11] motivate testing continuation against reversal, FinBERT follows Kirtac and Germano [2], and the allocations follow Colasanto et al. [7] and Coriat and Benhamou [6], with equal weighting as prior and benchmark [12].

Study	Data and method	Main finding
Tetlock [5]	Daily press column, word counts	Media pessimism predicts price pressure followed by reversal
Tetlock et al. [8]	Firm news, negative words	Unusual negative tone predicts lower earnings and returns
Heston and Sinha [9]	Firm news, daily vs. weekly	Weekly tone predicts returns for weeks, daily tone only for days
Ke et al. [10]	Newswires, supervised text model	News is priced within days, fastest for large firms
Hillert et al. [11]	2.2 million newspaper articles	Coverage amplifies momentum, which later reverses
Kirtac and Germano [2]	965,375 articles, FinBERT and LLMs	Transformer sentiment predicts returns better than dictionaries
Colasanto et al. [7]	BERT sentiment as Black–Litterman views	Sentiment views improved the optimised portfolio
Coriat and Benhamou [6]	Reinforcement learning with LLM sentiment	Combining price and sentiment beat either alone
DeMiguel et al. [12]	14 optimisers vs. $1/N$	Estimation error lets $1/N$ win out of sample

Table 1
Related work and its main findings.

These findings come mostly from U.S. long–short portfolios, single train–test splits or returns before costs. The gap is whether they hold inside one long-only, cost-aware optimiser for a transatlantic large-cap universe, with models selected on a separate validation period and news used only when available.

2. Theory

2.1. Market efficiency and investor psychology

Under semi-strong efficiency, a headline everyone can read should not predict returns [3]. In behavioural models, investors under-react to new information, so prices drift with the news, and over-react to a series of similar news, so prices later reverse [4]. If sentiment matters, returns after an unusual shift in tone should continue or reverse.

2.2. Measuring sentiment

FinBERT is a BERT language model adapted to financial text [1]. Its financial training texts (Reuters news from 2008–2010 and a 2014 bank of single news sentences, close in length to headlines) predate the test period, unlike those of newer large language models. For each headline it gives positive, negative and neutral probabilities, and tone is $s = p^+ - p^- \in [-1, 1]$. A persistent level of tone is not news, so, following Tetlock et al. [8], a firm's weekly mean tone $\bar{s}_{i,t}$ from $n_{i,t}$ headlines is compared with the mean $\mu_{i,t}$ and standard deviation $\sigma_{i,t}$ of its own weekly tone over the previous 52 weeks,

$$a_{i,t} = \frac{n_{i,t}}{n_{i,t} + k} \cdot \frac{\bar{s}_{i,t} - \mu_{i,t}}{\sigma_{i,t}}, \tag{1}$$

where $n/(n+k)$, with $k = 5$, shrinks the score when few headlines support it ($a = 0$ without news). The same measure over all headlines of a sector gives sector sentiment. The weekly scores are smoothed into an exponentially weighted average,

$$M_{i,t} = (1 - \lambda) M_{i,t-1} + \lambda\, a_{i,t}, \tag{2}$$

where $\lambda = 1 - 2^{-1/2} \approx 0.29$ gives a half-life of two weeks. Divided by its typical size across firms, M becomes the *sentiment gauge* G, measured in standard deviations. A view opens only when $|G|$ passes an entry band and closes inside a narrower exit band; this hysteresis limits trading [13].

2.3. Forecasting stock returns from prices

Besides linear regression, three families of price-based forecasts are common: time-series models (ARIMA); factor composites of 12-month momentum [14], one-month reversal and low volatility [13]; and tree ensembles and neural networks such as

LSTM, the strongest machine-learning methods for stock returns [15]. Forecasts are judged by the rank information coefficient (IC), the weekly rank correlation across stocks between forecast and realised return. It measures what the optimiser uses, the ordering of stocks [13], whereas squared errors are dominated by noise in weekly returns.

2.4. Portfolio optimisation with views

Mean–variance optimisation trades expected return against risk [16], but estimation error makes it fragile, and equal weighting is hard to beat [12]. The Black–Litterman model therefore moves the returns implied by a reference portfolio only where the investor holds views, in proportion to their confidence [17], and scaling views by measured skill gives a signal without skill no weight [13].

2.5. Conceptual model and hypotheses

Figure 1 links the constructs. A hypothesis is supported if the estimate has the expected sign and its 95% interval excludes zero, and supported in direction only if just the sign agrees.

H1 Sentiment moves prices beyond price information: after the gauge crosses its upper (lower) band, abnormal returns over the next 20 trading days are positive (negative) (H1a), and the gauge predicts next-week returns after controlling for price predictors (H1b).

H2 Correcting the price forecast with sentiment raises net return (H2a) and reduces maximum drawdown (H2b) compared with the price-only forecast, because price signals confirmed by sentiment earn more than contradicted ones (H2c).

H3 The sentiment-driven allocation has a higher Sharpe ratio after costs than equal weighting, minimum variance and the index.

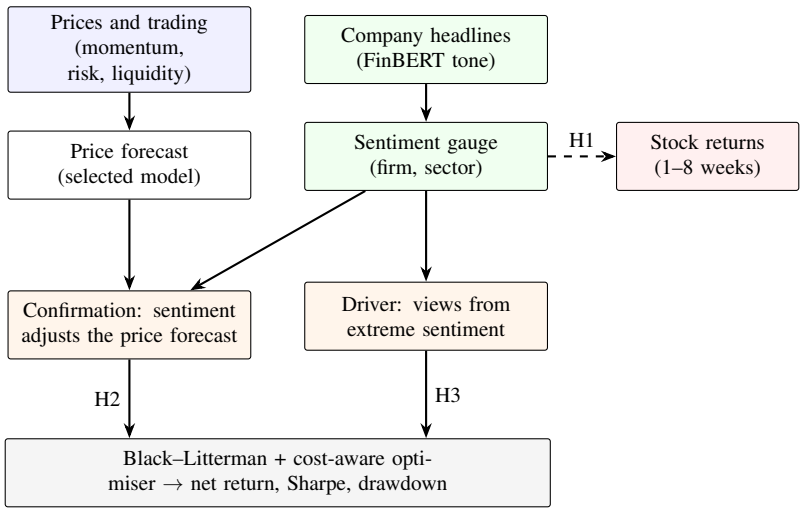


Fig. 1. Conceptual model. Price and news inputs form a price forecast and a sentiment gauge. Sentiment either confirms or contradicts the price-based forecast or drives the views itself. Labels show where each hypothesis is tested.

3. Data description

3.1. Sources and universe

All data are public. The universe is the members of the S&P 100 and the EURO STOXX 50 on 31 December 2014, the day before the test period, so membership is not chosen with hindsight. Of these 150 firms, nine lack a usable price history in the free data and are excluded: eight were acquired, merged or taken private between 2016 and 2025, and one has only a stale secondary listing. This leaves 141 firms, 92 in the United States and 49 in the euro area, in ten sectors (Appendix A). The exclusion uses later information; it affects 6% of the names, and the sign of the bias is unknown because takeover targets often gain when a bid is announced.

Daily prices, volume, exchange rates and the MSCI World benchmark (the URTH fund, with dividends) come from Yahoo Finance from 2005, and the risk-free rate is the three-month U.S. Treasury yield. Returns are in U.S. dollars, unhedged. Vivendi leaves the universe at its December 2024 break-up.

Headlines come from Google News searches on each company's name, one query per firm and calendar month from October 2010 to December 2025, all retrieved in October 2026.

3.2. Validity of the news data

Three properties of this source limit the backtest. First, and most important, the archive is retrospective. A backtest needs the headlines available at each decision, but a search run in 2026 returns what is still online, ranked by today's relevance. Removed articles are missing, the historical ranking may have differed, edited articles may carry later headlines, and stories that proved important afterwards may be over-represented while short-lived ones are under-represented. Because this favours news that mattered ex post, it is more likely to overstate than to hide predictive power. A check against 129,939 U.S. headlines dated from publishers' own archives (FNSPID [18]; Appendix C) found no such bias. Google's weekly headline counts follow that news flow and past price moves, but not future ones ($t = -0.17$).

Second, Google returns at most 100 items per query. The cap binds in 30% of test firm-months, more in the United States (38%) than in the euro area (15%) and more each year (17% in 2015, 56% in 2025), so heavily covered firms are sampled while thinly covered firms are nearly complete (tested in Section 6.3).

Third, only headlines are scored. They are short and can be ambiguous, unrepresentative of the article, repeated across publishers or written after the market has reacted. On 7,906 annotated financial headlines FinBERT's label was right in 73%, mostly confusing neutral with polar tone. Dropping the 5.0% of headlines that repeat a story in other words leaves price + sentiment unchanged but lowers the sentiment-driven Sharpe ratio to 0.57 (Appendix C).

3.3. Preparation and quality

Of 1,175,905 retrieved items, 661,986 company headlines remain (Figure 10, Appendix A). Identical headlines are removed, and a headline is kept only if it names the issuer. Names that are also common words or people, such as Apple or Ford, must appear together with business words. Market roundups naming three or more firms (1,789 headlines) are removed. Most headlines are close to neutral, while about 16% are clearly positive and 15% clearly negative ($|s| > 0.5$; Figure 11 in Appendix A). Euro-area headlines are more positive than U.S. headlines, and tone rose after 2020; such level differences are why tone is measured against each firm's own history. Google News reports only the publication date, so a headline counts only from the next opening of the issuer's exchange. Coverage is uneven across firms and sectors (Figure 12); the median firm has 4 headlines a week, and no news arrives in 19.7% of test stock-weeks, which are then assumed to carry neutral sentiment (tested in Section 6.3).

4. Methodology

4.1. Research design

The study is a quantitative simulation organised by the six phases of the Cross-Industry Standard Process for Data Mining (CRISP-DM) [19] (Figure 2). Decisions are weekly and walk-forward (Section 4.5), and all strategies share one optimiser, constraints and costs, so differences come from how sentiment enters the decision. All choices were made on the validation period, 2012–2014, then fixed with a cryptographic hash of the code and settings before the test period, January 2015 to December 2025 (573 weeks), was run once (Figure 3).

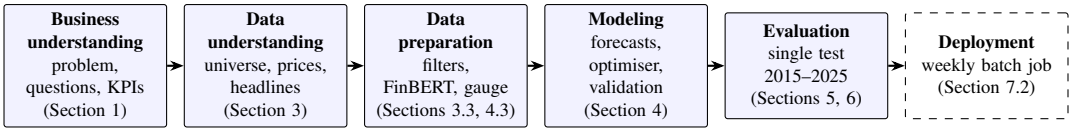


Fig. 2. Research design following the six CRISP-DM phases [19], with the sections that cover each phase. Deployment is discussed but not carried out (dashed).

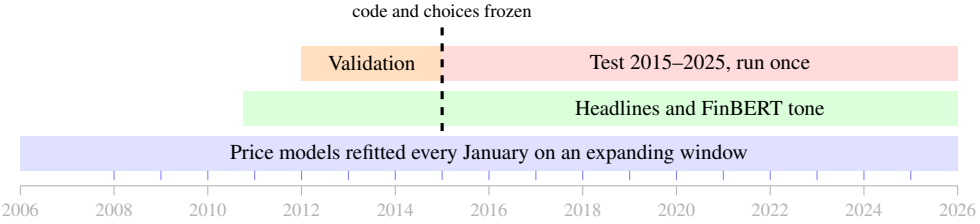


Fig. 3. Walk-forward timeline. Models are refitted every January (ticks) on data known before the first decision of the year; all choices are fixed on 2012–2014 before the test.

4.2. *Price forecasting models*

Six candidates represent these families: Ridge regression as the linear baseline [20], ARIMA per stock (order chosen by AIC), the factor composite, a random forest, gradient-boosted trees and an LSTM on the last eight weeks of inputs. The learning models use ten price and trading predictors (Table 2), ranked across stocks every week, and forecast returns relative to the cross-sectional mean. Fundamentals are not used because free point-in-time accounting data are unavailable for European firms. All models are pooled across stocks and refitted every January. The candidate with the highest mean weekly rank IC in the validation period is selected.

Group	Inputs (values known at the decision cutoff)
Price (7)	12-month momentum excluding the last month; 1-month and 1-week reversal; 60-day volatility; largest daily return over 20 days; beta to the universe over 52 weeks; distance to the 52-week high
Trading activity (3)	20-day dollar volume; illiquidity (mean absolute daily return per dollar traded); abnormal volume (20-day versus 120-day dollar volume)
Sentiment	abnormal firm tone and abnormal sector tone (Eq. 1), smoothed into the gauge

Table 2
Model inputs. Headline counts enter only through the shrinkage in Eq. (1).

4.3. *Sentiment gauge*

The gauge combines the smoothed firm and sector scores, $G = [(1-w)M^f + wM^s]/\text{s.d.}$, where the firm-only ($w = 0$) and sector-only ($w = 1$) versions are G^f and G^s. Validation chose between $w = 0$ and $w = 0.5$ by the larger absolute IC, and the sign of that IC decides whether the gauge is used for continuation or reversal. The s.d. is pooled across firms over the previous 52 weeks, which keeps the bands comparable as coverage grows. A view opens at $|G| \geq 1.5$ and closes below $|G| = 0.5$ (Figure 4).

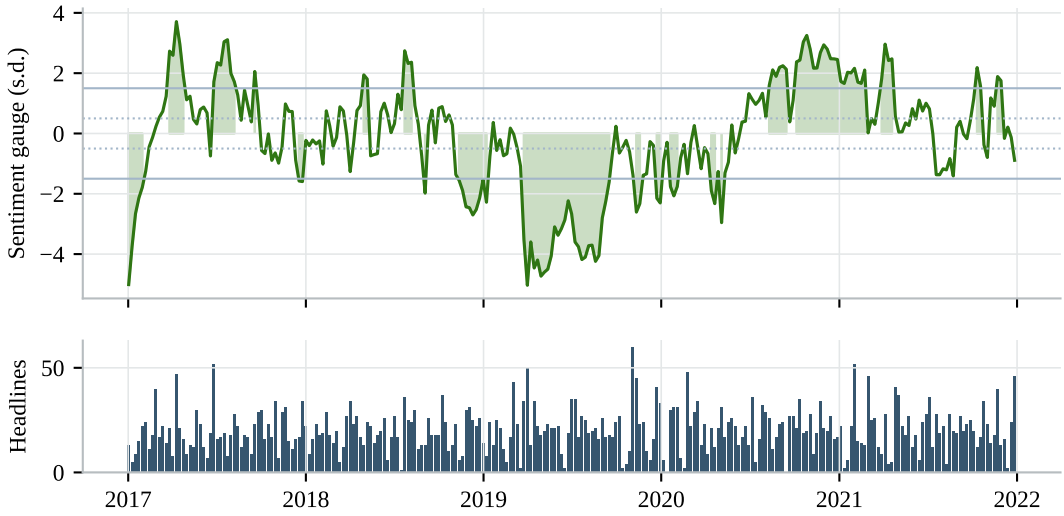


Fig. 4. Sentiment gauge and weekly headlines for Boeing, 2017–2021. Solid lines mark the entry bands and dotted lines the exit bands; shaded areas are weeks with the gauge beyond the entry band.

4.4. *Sentiment as confirmation of the price forecast*

The selected price forecast is first converted into an expected relative return $\hat{y}^P$ by a regression through the origin of realised relative returns on the forecast over the previous three years. A Ridge model then learns how sentiment corrects the forecast. To avoid look-ahead, it is trained only on *out-of-sample* errors $e = y - \hat{y}^P$, from forecasts made with earlier data, and is refitted every January:

$$\begin{aligned} \hat{y}^C_{i,t} &= \hat{y}^P_{i,t} + \beta_0 + \beta^\top \mathbf{x}_{i,t}, \\ \mathbf{x}_{i,t} &= \big(\mathbf{g}_{i,t},\ d_{i,t}\,\mathbf{g}_{i,t}\big), \\ \mathbf{g}_{i,t} &= \big(G^f, G^s, a^f, a^s\big)_{i,t}, \end{aligned} \tag{3}$$

where G^f and G^s are the firm and sector gauges, a^f and a^s the current week's abnormal firm and sector tone (Eq. 1), and $d_{i,t} = \pm 1$ the direction of the price

forecast; a simpler version without the weekly tone was rejected on validation. The agreement terms $d\,\mathbf{g}$ let sentiment strengthen a price signal it confirms and weaken, or overturn, one it contradicts. The Ridge penalty is chosen by generalised cross-validation [20].

4.5. Optimisation model

All optimised strategies solve the same problem every week (Table 3). The decision variables are the weights $\mathbf{w}$ of the N_t investable stocks, which solve

$$\max_{\mathbf{w}} \; \underbrace{\mu^\top \mathbf{w}}_{\text{return}} - \underbrace{\frac{\delta}{2}\mathbf{w}^\top \Sigma \mathbf{w}}_{\text{risk}} - \underbrace{\frac{c}{H}\sum_i |w_i - \tilde{w}_i|}_{\text{trading cost}} \qquad (4)$$

$$\text{s.t.} \;\; \textstyle\sum_i w_i = 1, \qquad 0 \le w_i \le 0.05 \quad \forall i,$$

a convex quadratic programme, where $\tilde{\mathbf{w}}$ are last week's weights after price drift. A trade is paid once but earns over several weeks, so its cost c is spread over the expected holding time H.

The expected returns μ are the Black–Litterman posterior around the reference portfolio, equal weighting. Its implied returns $\pi = \delta\Sigma\,\mathbf{w}_{\mathrm{EW}}$ make $\mathbf{w}_{\mathrm{EW}}$ optimal when there are no views and no costs. A view states that stock i will beat π_i by $\alpha_i = \widehat{\mathrm{IC}}_t\,\sigma_i\,z_i$, where σ_i is its weekly volatility and z_i the signal in standard deviations. With prior uncertainty $\tau\Sigma$ and view uncertainty $\Omega = \tau\mathbf{D}_S$ [17, 21], τ cancels:

$$\mu = \pi + \Sigma_{\cdot S}\big(\Sigma_{SS} + \mathbf{D}_S\big)^{-1}\alpha_S, \qquad (5)$$

where S is the set of stocks with views, $\Sigma_{\cdot S}$ the covariance of all stocks with them and $\mathbf{D}_S$ their variances.

The *confidence* in a view enters through Ω, which makes each view as uncertain as the stock's own return in the prior, so a single uncorrelated view moves μ_i halfway from π_i to $\pi_i + \alpha_i$. The size of the view follows the signal's record, as $\widehat{\mathrm{IC}}_t$ is its mean weekly rank IC over the previous three years, set to zero if negative or if fewer than 52 weeks are known, so a signal without measured skill has no views.

The strategies differ only in their views: *price only* (all stocks; z is the cross-sectional z-score of the price forecast), *price + sentiment* (all stocks; Eq. 3) and *sentiment-driven* (only stocks with an open gauge view; $z = G$, capped at ± 3). *Minimum variance* solves Eq. (4) with $\mu = \mathbf{0}$. Benchmarks are equal weighting, rebalanced weekly, and buy-and-hold URTH. The rebalancing rule is the same every week:

1. At the cut-off, Thursday 22:00 UTC after the U.S. close, form the investable set (stocks with at least 40 of the last 52 weekly returns), Σ, the signals and $\widehat{\mathrm{IC}}_t$ from data published before it.
2. Compute μ (Eq. 5) and solve Eq. (4) from $\tilde{\mathbf{w}}$.
3. Trade to the solution at each stock's Friday close, selling stocks that leave the universe, and pay $c\sum_i |w_i - \tilde{w}_i|$.
4. Hold for one week; the drifted weights become $\tilde{\mathbf{w}}$.

Element	Specification	Setting
Decision variables	Weight w_i of each investable stock after the weekly trade	N_t = 140–141 stocks
Objective	Maximise expected return minus risk and trading-cost penalties, Eq. (4)	–
Risk term	$\frac{\delta}{2}\mathbf{w}^\top\Sigma\mathbf{w}$; Ledoit–Wolf covariance of the last 52 weekly USD returns [22]	$\delta = 2.5$
Trading-cost term	$\frac{c}{H}\sum_i \lvert w_i - \tilde{w}_i\rvert$ from the drifted weights $\tilde{\mathbf{w}}$	$c = 10$ bp, $H = 4$ weeks
Reference portfolio	Equal weight; implied returns $\pi = \delta\Sigma\mathbf{w}_{\mathrm{EW}}$	$w_i = 1/N_t$
Expected returns	Black–Litterman posterior, Eq. (5)	τ cancels
Views	$\alpha_i = \widehat{\mathrm{IC}}_t\,\sigma_i\,z_i$ on the stocks in S	gauge capped at ± 3
View confidence	$\Omega = \tau\,\mathrm{diag}(\Sigma_{SS})$; view size scaled by the trailing IC	3-year IC, $\geq$52 weeks, floored at 0
Constraints	Fully invested, long-only, issuer limit (UCITS 5%); no leverage, cash or sector limits	$\sum_i w_i = 1$, $0 \le w_i \le 0.05$
Rebalancing	Decide Thursday 22:00 UTC, trade at each stock's Friday close, hold one week	573 weekly decisions
Solver	Convex quadratic programme	Clarabel (cvxpy)

Table 3
Specification of the optimisation model shared by all optimised strategies; minimum variance sets $\mu = \mathbf{0}$.

4.6. Costs and evaluation

Every purchase and sale costs $c = 10$ basis points of traded value, close to the measured costs of large institutional trades [23]; 0, 5 and 20 basis points are tested. KPIs are compound annual growth (CAGR), volatility, the Sharpe and Sortino ratios (Treasury yield as risk-free rate), maximum drawdown, turnover and cost. Differences between strategies get 95% intervals from a paired bootstrap that resamples 13-week blocks to keep autocorrelation (2,000 draws) [24]; mean ICs and alphas use Newey–West (HAC) standard errors [25]. RQ1 is tested with an event study

of daily returns relative to the stock's regional average around tradable gauge signals, and with Fama–MacBeth regressions of next-week returns on the gauges with and without price controls [26]. Robustness tests change one setting at a time. Table 6, Figure 7, Section 6.3 and Appendix C report analyses added after the test; they use the frozen models and change no choice.

5. Results

5.1. Model selection on the validation period

No price model forecast weekly returns reliably in 2012–2014 (Table 4). The factor composite had the highest mean rank IC, +0.0090 ($t = 0.39$), and was selected. The firm + sector gauge had a higher IC than the firm-only gauge (+0.0125 against +0.0039) and a positive sign, so it was used for continuation. Adding the weekly abnormal tone to the sentiment correction raised its validation IC from +0.0090 to +0.0108 (gain $t = 0.06$), so that version was kept.

Model	Mean rank IC	HAC t	Weeks IC > 0 (%)
Ridge regression	-0.0040	-0.25	49.7
Factor composite	+0.0090	0.39	54.1
ARIMA	-0.0038	-0.39	47.1
Random forest	-0.0000	-0.00	52.2
Gradient boosting	+0.0018	0.20	49.7
LSTM	-0.0031	-0.13	47.1
Sentiment gauge, firm only	+0.0039	0.60	51.6
Sentiment gauge, firm + sector	+0.0125	0.98	55.4

Table 4
Validation results, 2012–2014 (157 weeks). Rank IC with Newey–West (HAC) t-statistics; selected models in bold.

5.2. Sentiment and future returns (RQ1)

Figure 5 follows 1,689 positive and 1,925 negative gauge signals. Most of the abnormal return comes *before* the signal can be traded, with +0.37% and -0.68% in the ten days up to day 0. After day 0, positive signals drift up by a further +0.18% over 20 trading days and then fall back (-0.36% from day 0 to day 40), while negative signals move by -0.07% over 20 days and then recover (+0.51%). Over 20 days both groups move in the expected direction, but by amounts small relative to the intervals that fade by day 40, so H1a is supported in direction only.

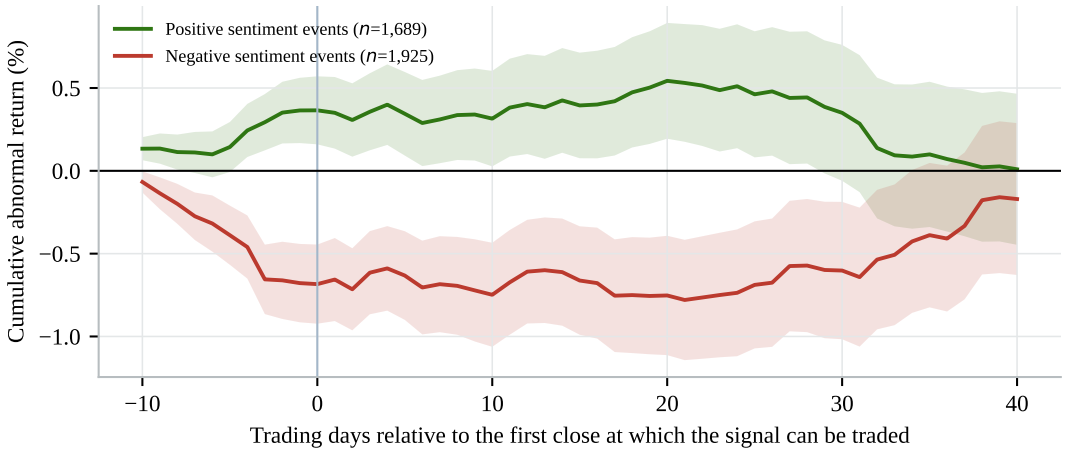


Fig. 5. Cumulative abnormal return (relative to the equal-weighted return of the stock's region) around gauge signals, 2015–2025, with 95% intervals. Day 0 is the first close at which the signal can be traded.

In Fama–MacBeth regressions (Table 5), a one-standard-deviation higher firm gauge predicts a +1.3 basis points higher relative return the next week after price controls ($t = 1.19$), +2.2 basis points in the United States ($t = 1.61$) and -0.9 in the euro area ($t = -0.45$). Sector sentiment is insignificant. H1b is supported in direction only for firm sentiment. Over horizons of one to eight weeks the gauge's rank IC is near zero, with its highest value, +0.0077 ($t = 1.11$), four weeks ahead (Figure 6).

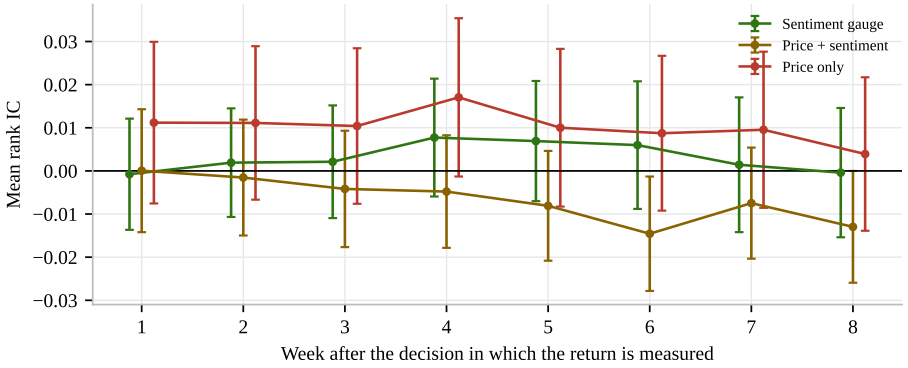


Fig. 6. Mean rank IC of each signal for returns one to eight weeks after the decision, with 95% intervals.

Regressor	Sentiment only	With price controls	U.S.	Euro area
Firm sentiment gauge	+1.1 (0.97)	+1.3 (1.19)	+2.2 (1.61)	-0.9 (-0.45)
Sector sentiment gauge	-1.9 (-0.79)	-0.8 (-0.41)	-1.1 (-0.51)	-0.9 (-0.37)
Momentum 12-1		+1.5 (0.55)	+2.5 (0.83)	+2.3 (0.72)
Reversal (1 month)		+4.9 (1.59)	+5.6 (1.47)	+3.5 (1.24)
Volatility (60 days)		-3.4 (-0.99)	-2.5 (-0.69)	-2.4 (-0.67)
Dollar volume		+3.3 (1.22)	+3.5 (1.63)	-3.9 (-1.94)
Beta (52 weeks)		+7.5 (1.49)	+4.7 (0.87)	+7.2 (1.58)
Mean R^2	0.036	0.242	0.271	0.326

Table 5
Fama–MacBeth regressions of next-week relative returns on standardised predictors, 2015–2025. Coefficients in basis points per standard deviation, Newey–West t-statistics in parentheses.

5.3. *Sentiment as confirmation of the price forecast (RQ2)*

In the test period the factor composite kept a small positive IC, +0.0112 ($t = 1.17$), while the corrected forecast fell to +0.0000 ($t = 0.01$) and the gauge itself had an IC of -0.0008 ($t = -0.12$). Table 6 shows the hit rates behind this result. The price signal and the sentiment gauge were each right about half of the time, whether they agreed or disagreed. When the correction moved a stock to the other side of the average (31% of cases), the original price decision would have been right more often than the new one (51.5% against 48.5%).

A. Did each signal pick the right side of the average? Situation	Share of cases	Price signal right	Sentiment right
Price and sentiment agree	31%	50.0%	50.0%
Price and sentiment disagree	28%	50.6%	49.4%
Sentiment neutral	41%	50.2%	49.9%
All cases	100%	50.2%	49.6%
B. Decisions changed by the sentiment correction	Share of cases	Original price decision right	Corrected decision right
Correction moved the stock to the other side	31%	51.5%	48.5%

Table 6
Who was right? One case is one stock in one week, 2015–2025. A signal is right when the stock it favours (penalises) does better (worse) than the average stock the next week; a coin flip is right 50% of the time. Sentiment agrees or disagrees when its gauge is at least 0.5 s.d. on the same or the opposite side of the price signal; cases with a gauge of exactly zero are left out of "Sentiment right".

The portfolio results are better than the forecast results (Table 7). The price + sentiment portfolio earned 12.20% a year with a Sharpe ratio of 0.65 and a maximum drawdown of -30.4%, against 10.28%, 0.54 and -34.4% for price only; the difference, +1.92 percentage points a year, has a 95% interval of [-1.47, 5.00] (Table 8). Figure 7 shows where the difference arises. As a simple average, the gain of +1.70 percentage points

a year splits into +1.52 before costs and +0.17 from lower costs. The before-cost gain came equally from the price portfolio's buy decisions (stocks it overweighted; +0.76 percentage points a year, $t = 0.70$) and sell decisions (stocks it underweighted; +0.77, $t = 0.95$). When sentiment enlarged a buy or a sell, the stock moved the intended way in 52.8% and 51.9% of cases. H2a and H2b are supported in direction but not statistically, and H2c in direction only: stocks moved +0.8 basis points a week in the price signal's direction when sentiment agreed and -0.9 when it disagreed (both insignificant).

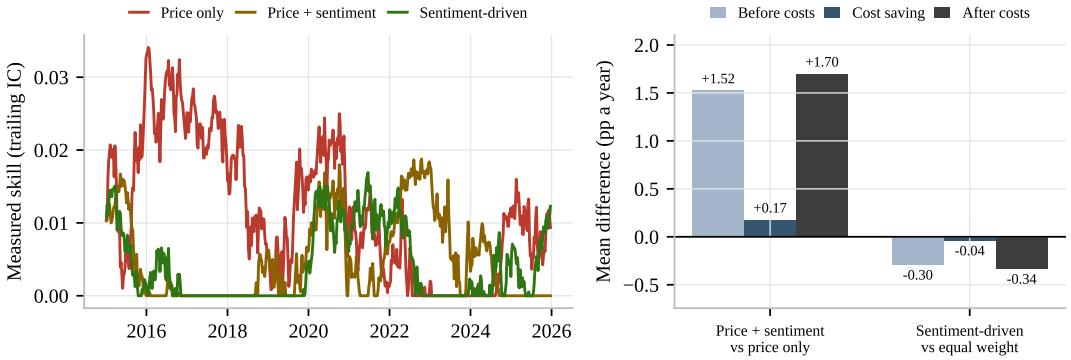


Fig. 7. Left: measured skill (trailing three-year rank IC, floored at zero) that scales each signal's views. Right: mean annual return difference before costs, from lower costs and after costs (arithmetic, percentage points).

5.4. *Sentiment as the driver of the allocation (RQ3)*

Driven by sentiment alone, the portfolio held about 54 stocks with views on 20 of them in an average week; in 45% of the weeks the gauge's trailing IC was zero and the optimiser moved towards equal weighting. The portfolio earned 11.90% a year, slightly less than equal weighting (12.15%), but with a Sharpe ratio of 0.64, against 0.63 for equal weighting, 0.60 for minimum variance and 0.60 for MSCI World (Figure 8; drawdowns and annual returns in Appendix B). Its maximum drawdown, -31.6%, was smaller than that of equal weighting (-36.7%) and the index (-32.7%). All differences have bootstrap intervals that include zero (Table 8). H3 is supported in direction only.

Strategy	CAGR	Vol.	Sharpe	Sortino	Max DD	Turnover	Costs	α (t)	β
Sentiment-driven	11.90	16.77	0.64	0.89	-31.6	1.6	0.17	+1.59 (+0.67)	0.91
Price + sentiment	12.20	16.86	0.65	0.93	-30.4	3.5	0.36	+1.62 (+0.81)	0.93
Price only	10.28	16.99	0.54	0.76	-34.4	5.2	0.54	-0.11 (-0.05)	0.94
Minimum variance	9.41	13.05	0.60	0.81	-28.2	0.5	0.05	+1.15 (+0.55)	0.67
Equal weight	12.15	17.46	0.63	0.90	-36.7	1.2	0.13	+1.30 (+0.65)	0.97
MSCI World (URTH)	11.18	16.59	0.60	0.85	-32.7	–	–	–	–

Table 7
Net performance, January 2015 to December 2025 (573 weeks), 10 bp per trade. CAGR, volatility, costs and alpha in % a year; turnover is the average weekly one-way turnover in %; alpha and beta against MSCI World with Newey–West t-statistics.

Comparison (A vs B)	Return (pp a year)	Sharpe ratio	Drawdown (pp)	Significant?
Price + sentiment vs price only	+1.92 (-1.47 to +5.00)	+0.10 (-0.09 to +0.28)	+4.0 (-5.5 to +9.5)	No
Sentiment-driven vs equal weight	-0.25 (-3.23 to +2.72)	+0.01 (-0.15 to +0.14)	+5.1 (-3.9 to +9.1)	No
Sentiment-driven vs minimum variance	+2.49 (-3.09 to +8.62)	+0.03 (-0.32 to +0.34)	-3.3 (-13.9 to +2.7)	No
Sentiment-driven vs MSCI World	+0.72 (-4.01 to +5.76)	+0.03 (-0.24 to +0.29)	+1.1 (-6.2 to +10.4)	No
Price + sentiment vs equal weight	+0.06 (-2.41 to +2.56)	+0.02 (-0.12 to +0.14)	+6.3 (-4.0 to +10.7)	No
Price only vs equal weight	-1.86 (-5.20 to +1.43)	-0.09 (-0.26 to +0.08)	+2.2 (-5.0 to +6.8)	No

Table 8
Are the differences real? Each cell shows strategy A minus strategy B (for drawdown, how much smaller A's drawdown was) with its 95% confidence interval in brackets (paired block bootstrap, 13-week blocks, 2,000 draws). A difference is significant only if its interval excludes zero.

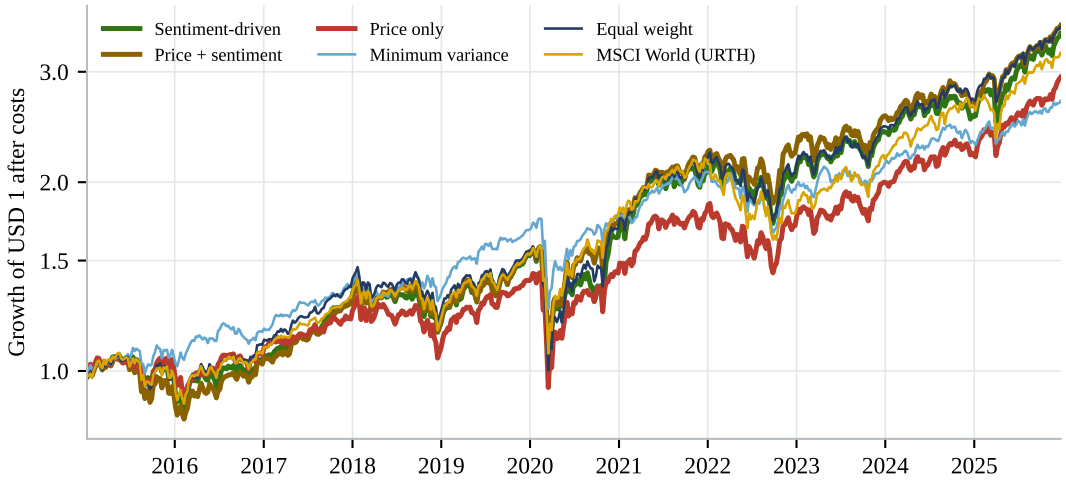


Fig. 8. Growth of USD 1 after costs, 2015–2025 (log scale).

6. Analysis

6.1. Sensitivity to costs, constraints and timing

In Table 9, the sentiment-driven portfolio's Sharpe ratio stays above equal weighting with a lower entry band of 1 s.d. (0.67), with a one- or four-week half-life, and when news reaches the portfolio one session later (0.66). The higher 2 s.d. band and the 10% position limit leave it slightly below.

The cost results are not monotonic (Figure 9). The sentiment-driven portfolio has a higher Sharpe ratio than equal weighting at every cost level, and without costs it earns 13.67% a year against 12.29%. At 20 bp all optimisers trade less, and the price-only portfolio performs best of all. A different cost rate also changes the trades, and the spread across scenarios is as large as the differences between strategies.

Scenario	Sentiment-driven	Price + sentiment	Price only	Minimum variance	Equal weight
Base case	11.90 / 0.64	12.20 / 0.65	10.28 / 0.54	9.41 / 0.60	12.15 / 0.63
Costs 20 bp per trade	12.99 / 0.70	12.72 / 0.70	13.74 / 0.75	10.41 / 0.66	12.00 / 0.62
Position limit 10%	11.73 / 0.62	12.05 / 0.64	10.67 / 0.56	8.80 / 0.56	12.15 / 0.63
Entry band 1 s.d.	12.57 / 0.67	–	–	–	–
Entry band 2 s.d.	11.48 / 0.61	–	–	–	–
Gauge half-life 1 week	12.70 / 0.68	12.30 / 0.65	–	–	–
Gauge half-life 4 weeks	12.13 / 0.65	13.02 / 0.70	–	–	–
News one session later	12.29 / 0.66	12.40 / 0.65	–	–	–
2015-2019	8.79 / 0.62	8.94 / 0.62	6.84 / 0.50	10.87 / 0.93	9.20 / 0.65
2020-2025	14.57 / 0.66	15.01 / 0.68	13.25 / 0.59	8.21 / 0.42	14.67 / 0.64

Table 9
Robustness: CAGR (%) / Sharpe ratio after costs. One setting is changed at a time; "–" means the strategy is unaffected. Costs of 0 and 5 bp are shown in Figure 9.

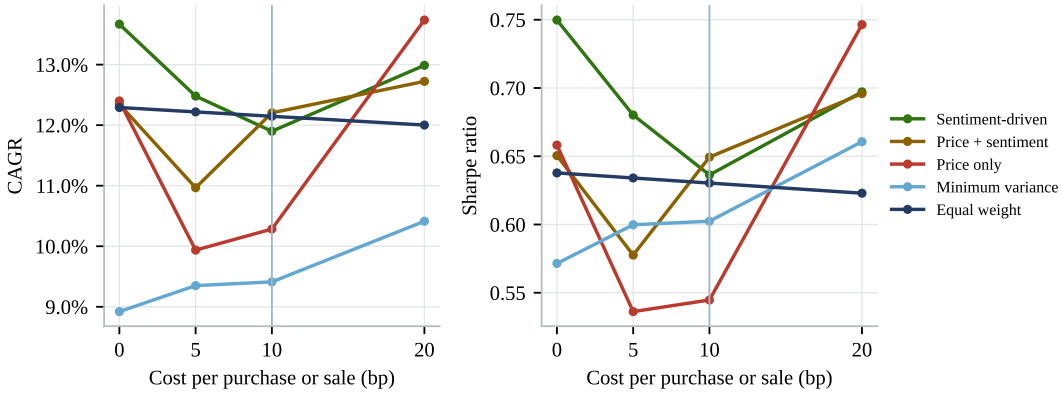


Fig. 9. Net CAGR and Sharpe ratio as functions of the cost per purchase or sale. The vertical line marks the base case.

6.2. Market conditions and sources of return

In 2015–2019 minimum variance had the highest Sharpe ratio (0.93) and the sentiment-driven portfolio trailed equal weighting slightly; in 2020–2025 the sentiment-driven and price + sentiment portfolios had the highest Sharpe ratios (0.66 and 0.68). Relative to equal weighting, the sentiment-driven portfolio returned -4.6 basis points a week in rising markets and +5.0 in falling markets (both insignificant), suggesting that its smaller drawdown comes from defensive behaviour in sell-offs.

On average the sentiment-driven portfolio held about 8 percentage points more information technology and 7 points less financials than equal weighting. Technology outperformed during the test period, so part of the result may come from this sector tilt rather than from picking stocks within sectors.

6.3. News coverage

Coverage is a firm's mean number of headlines a week over the previous 52 weeks; firms at or above the weekly median (3.8) are high-coverage, and the cap binds mainly for them.

High-coverage firms carried 68% of the sentiment views, but the gauge had no measurable skill in either group (Table 10). For the sentiment-driven portfolio, views only on high- or only on low-coverage firms, or none below one headline a week, gave Sharpe ratios of 0.64 to 0.70. The price + sentiment gain, in contrast, comes from heavily covered firms, where the gauge rests on more headlines. Correcting only these firms kept a Sharpe ratio of 0.65, while correcting only low-coverage firms gave 0.58, close to price only. The firm gauge's Fama–MacBeth effect was +1.5 against -0.1 basis points. All variants lie within the bootstrap interval of their base case. Stock-weeks without news earned +0.1 basis points relative to the average stock the next week ($t = 0.03$), consistent with treating them as neutral.

	High coverage	Low coverage
Share of sentiment views	68%	32%
Gauge rank IC (t)	-0.0002 (-0.03)	+0.0003 (0.04)
Return in view direction, bp/week (t)	-0.3 (-0.09)	+1.6 (0.22)
Contribution vs equal weight, pp/year (t)	+0.54 (0.69)	-0.84 (-0.75)
Firm gauge, Fama–MacBeth, bp (t)	+1.5 (1.00)	-0.1 (-0.05)
Sentiment-driven portfolio with views restricted by coverage		
Variant	CAGR (%) / Sharpe	Views per week
Base case	11.90 / 0.64	20.2
Views only for high-coverage firms	12.50 / 0.67	13.3
Views only for low-coverage firms	13.04 / 0.70	6.9
No views below 1 headline a week	11.97 / 0.64	18.1
Price + sentiment portfolio with the correction restricted by coverage		
Base case	12.20 / 0.65	–
Correction only for high-coverage firms	12.07 / 0.65	–
Correction only for low-coverage firms	10.82 / 0.58	–
No correction below 1 headline a week	12.29 / 0.65	–

Table 10
News coverage, 2015–2025 (post-test diagnostic). High-coverage firms have at least the weekly median number of headlines over the previous 52 weeks. Contribution is the return of the group's weights relative to equal weighting before costs, including stocks sold to fund views; the Fama–MacBeth row uses the price controls of Table 5; Newey–West t-statistics. Each variant changes one setting of the portfolio named in its panel; in the price + sentiment variants the other firms keep the price-only forecast.

7. Discussion

7.1. Interpretation and comparison with earlier studies

RQ1. Headline sentiment in large caps mostly confirms price moves that have already happened, which fits Tetlock's temporary price pressure [5] and the over-reaction to heavily covered news reported by Hillert et al. [11] better than the slow absorption of weekly tone found by Heston and Sinha [9]. The difference is consistent with Ke et al. [10], who find that news about large firms is priced within days. The firms here are heavily covered, and date-only news reaches the portfolio a day or more after professional feeds.

RQ2. Like Coriat and Benhamou [6], combining price and sentiment beat the price-only allocation in return, Sharpe ratio and drawdown. Unlike in their study, the correction did not make the forecast more accurate, and the gain is not statistically significant. The gain is economically relevant. In liquid large caps a credible edge is small and spread over many decisions [13], and 1–2 percentage points a year, earned on both buy and sell decisions, matters to an investor, although eleven years are too short to separate it from luck.

RQ3. Using sentiment as Black–Litterman views, as Colasanto et al. did [7], gave a marginally higher Sharpe ratio and a smaller drawdown than the benchmarks, with little trading. DeMiguel et al.'s warning that $1/N$ is hard to beat [12] holds here as well, since no difference is statistically significant over eleven years. A drawdown five percentage points smaller than equal weighting at a similar return is valuable to an investor who must hold through sell-offs.

7.2. Implications for portfolio managers

Two guidelines follow. First, Google News headline sentiment is better used with a price-based forecast than as a stand-alone source of alpha in liquid large caps. Second, any signal should enter through a skill-weighted, cost-aware optimiser, which limits the damage when it stops working. Deployment, the last CRISP-DM phase, would be a weekly batch job that runs in minutes on a laptop.

7.3. Limitations and future work

The main limitation is the news archive (Section 3.2): retrieved in 2026, it may lack removed articles and favour stories that later proved important. A check against dated U.S. headlines found no sign of this but cannot detect articles removed from both sources. Only headlines were scored, the sentiment-driven result depends on how repeated headlines are counted, and the universe excludes nine firms that later disappeared, a bias of unknown sign. Many comparisons were made, eleven years give little power to detect Sharpe differences of about 0.1, the cost penalty was not tuned, and MSCI World also contains markets outside the universe.

Future work should repeat the study on a historical, time-stamped news archive or commercial feed that records publication time, headline, source, company tags, corrections and updates, and de-listing and ticker history; GDELT, recorded as news has been published since 2015, is a free option [27]. This would remove

the retrieval bias, make coverage consistent and allow horizons shorter than a week. A mid-cap universe and a cost penalty chosen on validation are further extensions.

8. Conclusion

This study asked how Google News headline sentiment, scored by FinBERT, can improve a weekly, long-only U.S. and euro-area allocation after costs.

RQ1. Sentiment signals mostly followed price moves; the small continuation over the next month faded by day 40 (H1a in direction only). Firm sentiment had a small, insignificant positive effect after price controls (H1b in direction only).

RQ2. Correcting the factor forecast with sentiment did not raise its IC, but the portfolio earned +1.92 percentage points more a year with a smaller drawdown (H2a–H2c in direction only).

RQ3. The sentiment-driven allocation earned 11.90% a year, slightly below equal weighting, with a Sharpe ratio of 0.64, marginally above equal weighting (0.63), minimum variance (0.60) and MSCI World (0.60), and a smaller drawdown than equal weighting and the index. The differences are not significant and depend on costs, period and how repeated headlines are counted (H3 in direction only).

For a portfolio manager, Google News headline sentiment is useful as a confirmation layer for a price-based allocation and weak as a source of alpha on its own, although no improvement was statistically significant. Because the headlines were retrieved in 2026 and may favour stories that stayed online or later proved important, this should be confirmed on a point-in-time news archive before use.

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Appendix A. Universe and news coverage

Table 11 lists the firms with their accepted headlines for October 2010–December 2025. Figure 10 shows how the retrieved items are filtered, Figure 11 the distribution and regional level of FinBERT tone, and Figure 12 the average number of headlines per firm and week by sector and year.

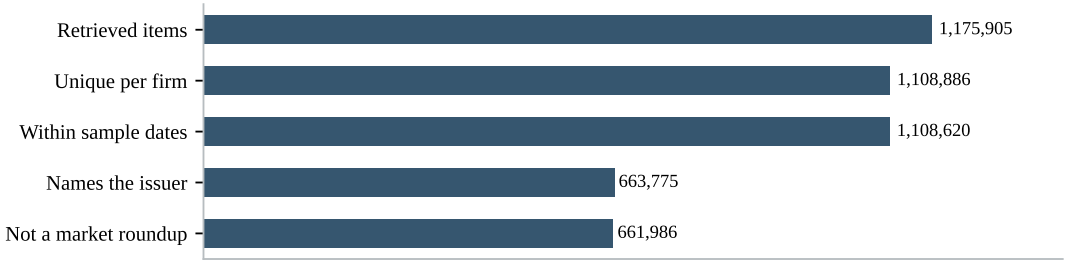


Fig. 10. Headline retention from retrieval to the weekly decision panel.

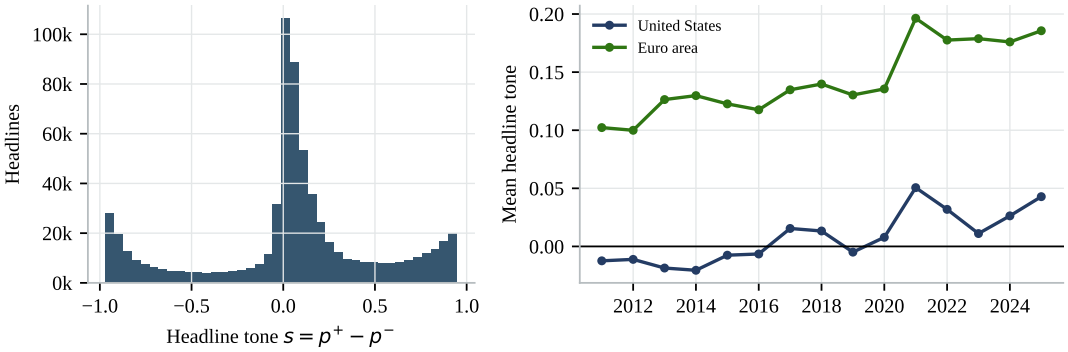


Fig. 11. FinBERT tone of the accepted headlines: distribution (left) and annual mean by region (right).

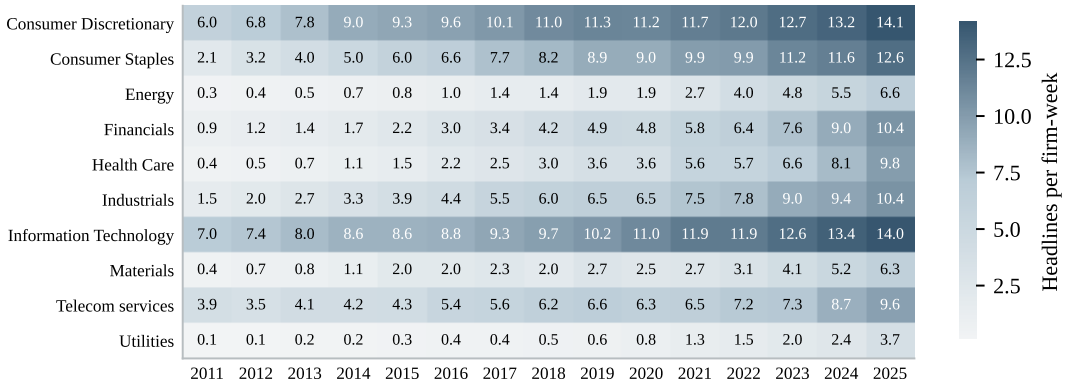


Fig. 12. Accepted headlines per firm and week, by sector and year.

Appendix B. Drawdowns and annual returns

Figures 13 and 14 add the path behind the drawdowns and returns in Table 7: when each strategy lost value, and how its return varied from year to year.

Ticker	Company	Headlines	Ticker	Company	Headlines
BMW.DE	BMW	16,585	COP	ConocoPhillips	957
ITX.MC	Inditex	856	CVX	Chevron	3,815
MBG.DE	Daimler (Mercedes-Benz Group)	4,070	DVN	Devon Energy	260
MC.PA	LVMH	4,321	HAL	Halliburton	1,255
VIV.PA	Vivendi	468	NOV	National Oilwell Varco	89
VOW3.DE	Volkswagen	16,019	OXY	Occidental Petroleum	1,291
ABI.BR	Anheuser-Busch InBev	3,395	SLB	Schlumberger	1,936
BN.PA	Danone	2,057	XOM	Exxon Mobil	3,949
CA.PA	Carrefour	2,226	AIG	American International Group	3,362
OR.PA	L'Oreal	6,314	ALL	Allstate	1,912
ULVR.L	Unilever	6,547	AXP	American Express	6,054
ENI.MI	Eni	2,009	BAC	Bank of America	7,448
REP.MC	Repsol	1,394	BK	Bank of New York Mellon	1,475
TTE.PA	Total (TotalEnergies)	4,271	BRK-B	Berkshire Hathaway	3,304
ALV.DE	Allianz	2,134	C	Citigroup	6,899
BBVA.MC	BBVA	2,110	COF	Capital One	4,181
BNP.PA	BNP Paribas	2,844	GS	Goldman Sachs	9,534
CS.PA	AXA	2,032	JPM	JPMorgan Chase	8,863
DBK.DE	Deutsche Bank	4,626	MET	MetLife	3,251
G.MI	Generali	803	MS	Morgan Stanley	5,510
GLE.PA	Societe Generale	1,114	SPG	Simon Property Group	252
INGA.AS	ING Groep	390	USB	U.S. Bancorp	3,685
ISP.MI	Intesa Sanpaolo	455	WFC	Wells Fargo	7,817
MUV2.DE	Munich Re	815	ABBV	AbbVie	2,015
SAN.MC	Banco Santander	2,607	ABT	Abbott Laboratories	2,653
UCG.MI	UniCredit	935	AMGN	Amgen	1,964
BAYN.DE	Bayer	2,756	BAX	Baxter International	1,692
EL.PA	Essilor (EssilorLuxottica)	284	BIIB	Biogen	1,378
SAN.PA	Sanofi	3,112	BMY	Bristol-Myers Squibb	1,219
AIR.PA	Airbus	11,915	GILD	Gilead Sciences	2,642
DG.PA	Vinci	757	JNJ	Johnson & Johnson	6,174
DHL.DE	Deutsche Post DHL	5,565	LLY	Eli Lilly	4,391
PHIA.AS	Philips	2,641	MDT	Medtronic	3,135
SGO.PA	Saint-Gobain	775	MRK	Merck & Co.	3,991
SIE.DE	Siemens	8,343	PFE	Pfizer	6,621
SU.PA	Schneider Electric	2,080	UNH	UnitedHealth Group	1,720
ASML.AS	ASML	1,640	BA	Boeing	15,084
NOKIA.HE	Nokia	10,317	CAT	Caterpillar	2,539
SAP.DE	SAP	3,384	EMR	Emerson Electric	245
AI.PA	Air Liquide	584	FDX	FedEx	9,757
BAS.DE	BASF	2,041	GD	General Dynamics	1,542
DTE.DE	Deutsche Telekom	624	GE	General Electric	6,682
ORA.PA	Orange	70	HON	Honeywell	4,170
TEF.MC	Telefonica	1,130	LMT	Lockheed Martin	8,471
ENEL.MI	Enel	794	MMM	3M	2,017
ENGI.PA	GDF Suez (Engie)	999	NSC	Norfolk Southern	1,595
EOAN.DE	E.ON	575	RTX	United Technologies (RTX)	468
IBE.MC	Iberdrola	943	UNP	Union Pacific	2,180
RWE.DE	RWE	1,099	UPS	United Parcel Service	3,417
AMZN	Amazon.com	7,347	AAPL	Apple	15,758
CMCSA	Comcast	6,589	ACN	Accenture	5,115
DIS	Walt Disney	15,723	CSCO	Cisco Systems	7,165
F	Ford Motor	6,189	EBAY	eBay	12,534
GM	General Motors	8,799	GOOGL	Google (Alphabet)	13,809
HD	Home Depot	6,777	HPQ	Hewlett-Packard (HP Inc)	1,182
LOW	Lowe's	3,021	IBM	IBM	9,164
MCD	McDonald's	15,468	INTC	Intel	14,687
NKE	Nike	6,195	MA	Mastercard	6,149
SBUX	Starbucks	14,342	META	Facebook (Meta Platforms)	16,273
TGT	Target	6,184	MSFT	Microsoft	12,010
CL	Colgate-Palmolive	818	ORCL	Oracle	5,346
COST	Costco	10,795	QCOM	Qualcomm	6,955
CVS	CVS Health	7,747	TXN	Texas Instruments	800
KO	Coca-Cola	13,440	V	Visa	1,794
MDLZ	Mondelez International	2,436	DD	DuPont	4,819
MO	Altria	716	FCX	Freeport-McMoRan	436
PEP	PepsiCo	11,284	T	AT&T	11,195
PG	Procter & Gamble	4,939	VZ	Verizon	10,460
PM	Philip Morris International	1,825	EXC	Exelon	437
WMT	Walmart	16,087	SO	Southern Company	404
APA	Apache	87			

Table 11
Research universe: members of the S&P 100 and EURO STOXX 50 on 31 December 2014 with usable price histories, and their accepted headlines.

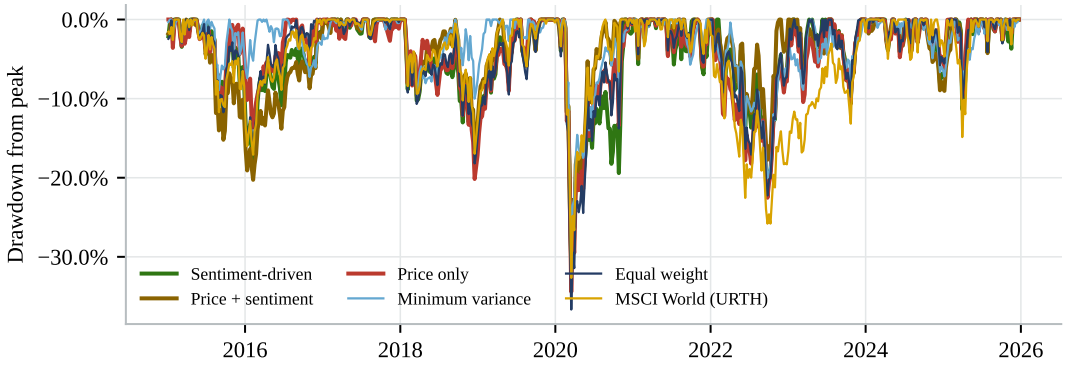


Fig. 13. Drawdown from the previous peak after costs, 2015–2025.

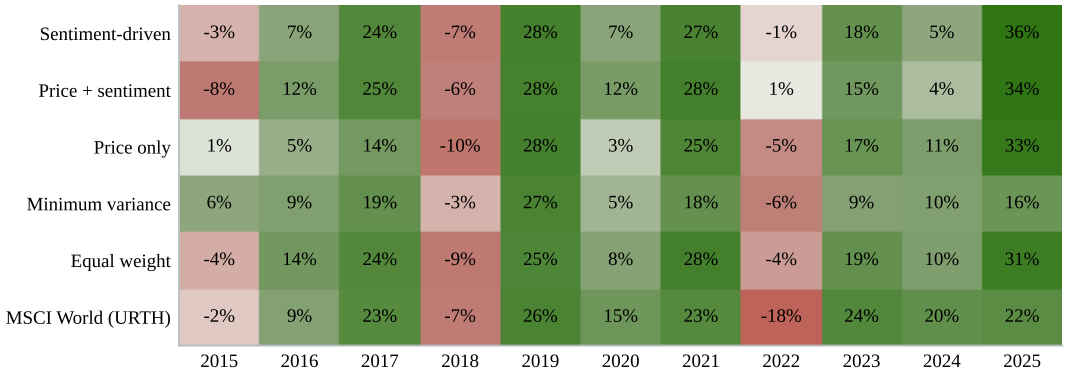


Fig. 14. Net annual returns by strategy and year.

Appendix C. Checks of the news data

C.1. Dated headlines

FNSPID [18] merges Nasdaq.com news with an external archive, mainly Benzinga, and dates each headline from its publisher's archive without search ranking or a cap per query. It covers U.S. firms to 2023 and was itself collected in 2023, so it cannot reveal articles removed from both sources. After the same issuer and roundup filters it holds 129,939 headlines for the 92 U.S. firms. Of 3,095 headlines found in both sources, Google gives the same date for 82%, a later date for 11% and an earlier one for 7%. Table 12 tests whether Google's search favoured stories that later moved prices: if so, Google's headline count would rise with future absolute returns once the news flow recorded by FNSPID is controlled for, which the table does not show.

	FNSPID	Google News	
Regressor	(1)	(2)	(3)
FNSPID headlines (log)	–	–	+5.3 (12.57)
$\lvert r\rvert$, weeks -4 to -1	+4.8 (10.10)	+1.4 (4.24)	+1.1 (3.26)
$\lvert r\rvert$, week 0	+7.3 (16.52)	+2.8 (8.63)	+2.2 (6.86)
$\lvert r\rvert$, weeks $+1$ to $+4$	-0.1 (-0.30)	-0.1 (-0.30)	-0.1 (-0.17)
Firm-weeks / weeks		43,148 / 469	

Table 12
Does the 2026 search favour stories that later moved prices? Fama–MacBeth regressions of firm-demeaned log(1 + headlines) per firm and week on standardised regressors, U.S. firms, test weeks 2015–2023. Coefficients in % per standard deviation, Newey–West t-statistics in parentheses. $\lvert r\rvert$ is the absolute return relative to the average stock; week 0 ends at the first close after the decision. (1) FNSPID headlines; (2) and (3) Google headlines.

C.2. FinBERT on headlines

SEntFiN 1.0 [28] holds 10,753 financial news headlines in which annotators labelled the sentiment towards each named company; FinBERT was not trained on them. Table 13 compares FinBERT's label, scored as in this study, with the annotation for the 7,906 headlines that name one company. Accuracy is 73%, and most errors confuse neutral with positive or negative tone. Among headlines labelled positive or negative that FinBERT also called positive or negative, it chose the wrong sign in 7.5%, and when its tone was strong ($|s| > 0.5$) the annotators agreed with its direction in 83% of cases. The headlines come from Indian financial media, so the test measures how FinBERT handles short headlines rather than the study's exact sources.

	FinBERT label (% of row)			
Annotated label	Positive	Neutral	Negative	Headlines
Positive	70	23	7	2,837
Neutral	12	75	13	2,693
Negative	5	21	74	2,376
Accuracy 73%, macro F1 0.73; opposite sign in 7.5% of polar predictions				

Table 13
FinBERT against human annotation of single-company financial news headlines (SEntFiN 1.0). Rows: annotated label; columns: FinBERT's most likely label, % of the row.

C.3. Repeated headlines

The main data remove only identical headlines. A headline is counted as a repeat when an earlier headline about the same firm within three days shares at least half of its words, as in "Pfizer agrees to pay $93 mln to settle Lipitor antitrust lawsuit" and the same headline with "$93 million". Repeats are 5.0% (33,267) of the accepted headlines. Without them the gauge correlates 0.99 with the original. The price + sentiment portfolio is unchanged, 12.38% a year (Sharpe 0.66) against 12.20% (0.65) in the test, but the sentiment-driven portfolio falls to 10.79% (0.57) from 11.90% (0.64), below equal weighting; the Sharpe difference has a 95% interval of -0.14 to +0.01. With a stricter rule (70% of words shared, 1.6% of headlines) the two portfolios earn 12.24% (0.65) and 11.62% (0.62). Repeats mark stories that many outlets carried, so the number of headlines also measures attention. The sentiment-driven portfolio acts only in the weeks when the gauge crosses its band and is sensitive to this, while the correction uses the gauge in every week and is not. The comparison also shows headlines that name a firm without being about it, such as a basketball final at Oracle Arena; the issuer filter does not remove them.

Appendix D. Reproduction

All results, tables and figures are produced from the stored data by `project.py` in four stages: `validate-price` and `validate-sentiment` (2012–2014 only), `freeze`, which stores a SHA-256 fingerprint of the code, settings and validation choices, and `test`, which refuses to run if anything changed after freezing and runs only once. Package versions and input file hashes are recorded in `results/test_run.json`. The post-test checks in Section 6.3 and Appendix C come from `coverage.py`, `fnspid_check.py` and `news_quality.py` in `diagnostics/`, which use the frozen models and leave the fingerprint unchanged.